Cosmological Relativity: what the construction delivers, what it declines, and where its edge is

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Abstract

This paper states, in one place, what the Cosmological Relativity corpus establishes, at the weight each result is established at, together with what the construction declines to claim and where its open edge lies. Nothing here is new: every claim is drawn from one of seventeen papers or from the ledgers those papers cite, and is stated at the register its source gives it—a proposition as a proposition, a reading as a reading, a conjecture as a conjecture. What is new is the assembly, and the two things an assembly can supply that a set of papers cannot: the joins, where a result in one paper is load-bearing for a result in another, and the relative size of the frontier against the body of work that surrounds it.

1 What forces the object, what it is, and the distinctions it forces

Cosmological Relativity is a construction on one geometric object, and almost everything below is that object read from somewhere. But the object is not where the argument starts, and presenting it first would ask a reader to grant the thing in question. It is reached: a measurement forces a cosmic foliation, reading that foliation ontologically is the augmentation, and maximal symmetry then leaves exactly one manifold to carry it. This section takes those in order, and closes on the distinction every later result depends on — what a *boundary* of such an object is.

1.1 What the measurement forces

The cosmological redshift is not a property an emitting source carries. It is the path integral of the expansion rate along the photon's own trajectory, accumulated continuously rather than fixed at emission [10]. That distinction is what makes the isotropy of the microwave background's monopole a measurement of something other than what it is usually taken to measure.

Grant the common reading its strongest form. Let last scattering be *perfectly* homogeneous. Then the observed temperature contrast is exactly the variation in the integrated expansion, with nothing left over, and *the entire observed isotropy is the statement that the integrated expansion was the same in every direction.* Homogeneity at the source does not *supply* the isotropy; granted perfectly, it removes one term and leaves the isotropy as a direct measurement of uniform expansion [10].

Under the alternative — that the large-scale expansion rate tracks the matter lumpiness region by region, with no single global scale factor — the accumulated redshift would scatter across the sky by $\sim 10^{-3}$, against an observed $\lesssim 3 \times 10^{-6}$: *an exclusion by three orders of magnitude. And every choice in the estimate biases it downward, so the number is a floor rather than an estimate* [10].

Two escapes remain and they are closed separately. The *statistical* escape is closed by the floor. The *structured* escape — a finely tuned, spherically symmetric inhomogeneity centred on the observer, which would give an isotropic redshift with no global uniform expansion and is untouched by any $\sqrt{N}$ argument — is closed by the Copernican principle *together with* the independently measured isotropy of the expansion history itself. *With both excluded, the observed isotropy forces uniform cosmic expansion.*

What a single datum thereby establishes is worth stating exactly, because it is larger than a parameter bound. The rejected region is not a corner of parameter space but *the entire class of histories lacking a single global scale factor on a single global time.* To reject that class is to select a global, time-ordered expansion — the cosmic foliation, made dynamical. *And the foliation is logically prior to everything the standard model otherwise assumes*: "space" is one of its slices, "expansion" an ordered family of them, isotropy and homogeneity properties of a slice — none of them statable until the foliation is fixed.

1.2 The augmentation that follows, and its structural counterpart

Augmenting general relativity by fixing a physical foliation and reading it ontologically — the lapse the objective rate at which the existing world advances, the shift the relativity of synchrony — is *necessary and sufficient* for a coherent formal description of an existing, evolving world [11]. *Necessary*, because a description lacking the lapse part collapses existence into occurrence. *Sufficient*, because the two parts close the only coherent escapes. No field equation is altered.

And the same augmentation is forced a second time, from general relativity alone. The event horizon of a collapsing body is a metric singularity met only at infinite exterior time, so no finite layer carries a completed horizon (§1.6) [2]. *These two forcings are independent and co-equal, and neither rests on the other*: one is measured and one is structural, and a reader may take either without the other.

1.3 Three senses of "real", fixed before the word is used

The corpus uses *real* in three senses and keeps them apart; this paper does the same. They are not degrees of one thing.

Real in the arithmetic sense: a geometry has a real coordinate basis, with no imaginary coordinate doing any work. That is the sense in which the substrate below is a real manifold, and the claim is about coordinates and signature and nothing more.

Real by construction: a feature is built by a construction and really seen from the vantage that builds it. The perspectival mass is of this kind, and so is the compact face on which a continuous $\mathfrak{su}(3)$ could live.

Existent: the ontological sense, whose criterion in this framework is temporality. *The existent is the evolving three-dimensional layer, and the manifold is its record* [11]. By that criterion the compact face is real by construction but, being Riemannian and atemporal, *not a co-equal existent*—it holds no clock [12].

So nothing in this section claims that a five-dimensional object is what exists. The substrate is the geometric core the construction reads and cuts; what exists, on this framework's own criterion, is the layer. The higher-dimensional geometry is what the layer is a cut of, not a thing possessing the layer's kind of being, and the four-dimensional manifold is the record rather than the existent.

1.4 What maximal symmetry then selects

The foliation is forced and the manifold that carries it is not yet fixed, and this is where the construction's one object arrives — as a selection rather than a premise. The maximally symmetric solutions of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ form a one-parameter family with curvature radius $\alpha = \sqrt{3/|\Lambda|}$. They are usually met as hypersurfaces in a flat five-dimensional space, which is a convenient picture and not the manifold: solving that embedding for the fifth coordinate and substituting back returns an intrinsic four-dimensional line element in which the manifold is coordinatised by four *real* coordinates, real regardless of the sign of the curvature.

The imaginary fifth coordinate is a property of one embedding picture, not of the surface. *This is reality in the first of the three senses above, and in that one only.*

Reading the metric's eigenvalues off that real line element gives the result the programme's ontology stands on. Three eigenvalues are positive always, and the fourth's sign is fixed by real data alone.

Proposition 1 (The unique intrinsically Lorentzian maximally symmetric manifold)**.** *De Sitter space is the only real maximally symmetric manifold carrying an intrinsic Lorentzian signature. It has a real coordinate basis, satisfies the maximal-symmetry requirement, and is the only such manifold with the requisite causal structure [14, 1].*

Two ways the signature can turn are worth separating, because they are of different kinds. At fixed positive curvature the crossing occurs where the fourth eigenvalue passes through a *pole*: its magnitude diverges and its sign flips, the numerator being constant throughout. *So the metric does not degenerate at the crossing*—a signature change through zero would send the determinant to zero and leave no metric there, and that is not what happens. The one place the eigenvalue vanishes is the null cone: not a crossing within a member of the family, but the family's own singular leaf [14].

That the substrate is reached through imaginary instruments is therefore a fact about the instruments. The embedding coordinate, the equatorial seam's continuation, and the reassignment at the cosmogenesis are each places the construction uses an imaginary variable and lands on a real manifold; the geometry is intrinsically real at every point they land on [14]. In one case the instrument is not optional and the case is the construction's own central algebraic object: over the field of the mass parameter the horizon cubic is irreducible, and below the Nariai mass its discriminant is positive, so its three roots are real and distinct. That is exactly the hypothesis pair of *casus irreducibilis*, whose conclusion is that no root lies in any real radical extension—*the three real horizon radii admit no expression in real radicals as functions of the mass*, and every radical route to them passes through $\mathbb{C}$ [14]. The imaginary is the only available road, and the roots it lands on are real.

1.5 One scale, and what that means

The substrate carries a single dimensionful scale. Read projectively it is a Cayley–Klein geometry[L] whose *absolute* is the null cone, and its geodesic separation is the Cayley–Klein log-cross-ratio taken against that absolute, with the Cayley–Klein constant equal to α [14]. So α is not a length inserted into a metric but the constant of a projective geometry whose absolute is the light cone—which is the signature proposition above, read projectively rather than metrically.

The constants through which physics is written on it are unit gauges, each a nameable feature of the substrate: c is the null-ruling slope of the equilateral hyperboloid, Λ the waist, and G enters only as the offset-length GM/c^2, the mass being the offset of the cut from the central geodesic [14, 8, 4]. On this accounting the gravitational–cosmological–quantum sector spends no free dimensionless constant, and the reason is structural rather than counted: neither real form supplies a second invariant, and a dimensionless magnitude needs two [14].

What the count leaves is not a constant but an initial condition, and the corpus carries it as measured content rather than as a derived quantity—as flat ΛCDM carries the baryon-to-photon ratio from a baryogenesis it does not model [9].

1.6 What a boundary of the substrate is

Here the corpus draws the distinction that everything downstream depends on, and it is a distinction between two kinds of singular locus rather than between a real one and an artefact.

A *metric* singularity is a locus at which a spatial extent contracts to zero while the curvature stays finite. A curvature singularity is a locus at which an invariant of the curvature diverges.

The event horizon of a collapsing body is the first kind: a null hypersurface along whose generators the spatial extent also contracts to zero, occurring only as the null future boundary approached at infinite exterior time [2]. Three consequences of that follow immediately, and each is a distinction a reader needs before meeting any application.

Observer-dependence is not the criterion. It is tempting to sort these loci by whether they are seen by every observer, and the sorting fails at once: the Rindler horizon is perspectival and carries a flux, so "perspectival, therefore no flux" is refuted before it starts. What sorts the four horizons the corpus treats is *completion*—whether the boundary is reached on any finite layer—and completion sorts all four [2].

And the finite-curvature case is the one the metric-singularity result treats. It is a different object from the locus standardly called the curvature singularity at $r = 0$, where the invariants computed on the areal radius diverge; the first is established with the event horizon as its exemplar, and nothing is said there about the second beyond noting that it is a different object [2]. *That the two admit a common treatment, and what becomes of the divergence at $r = 0$ under it, is established in the companion* [3, 4]. *No claim of necessity or sufficiency is made here in either direction*: the scope is what is stated, and reading it as a general criterion for when a boundary may be crossed would assert what the source declines to assert.

And formal likeness does not sort the two classes. A chart such as Eddington–Finkelstein parametrises approaches to a singular locus by a coordinate that degenerates there, so a single metric place is drawn spread out as an extended line. Mercator draws the North Pole by the same projective mechanism, and the visual appearance is identical: a place drawn as a line. *But the reason for the appearance is the opposite.* At the pole the chart manufactures an extension that is not there, and a chart centred on the pole collapses the line back to the point it always was. At the horizon the chart spreads out a metric collapse that *is* there: reading the line back yields no ordinary point a better chart would reveal, because no chart removes it—*the collapse is in the metric, not in the projection* [3]. So no argument from how a singular locus is drawn can sort the two, in either direction.

1.7 What follows for the reading, and what does not

Two guards belong here rather than later, because both are misread in the same direction.

Perspectival does not mean unreal. Where the corpus calls a feature perspectival it means built by a construction and really seen from the vantage that builds it—the curvature divergence of a perspectival metric is real as that metric's, and reading it as an artefact in the pejorative sense is the misreading the epistemology is written to defeat [5, 3].

And identifying a feature as perspectival is not a dismissal of the construction that carries it. Sweeping a radial curve to recover a spatial geometry is the legitimate and indeed the natural way to chart a geometry from a given vantage, and the result agrees with the standard geometry on every observable outside the horizon. What is identified is not an illegitimate operation but a misreading of its product: treating features the construction manufactures—the asymmetric labelling, and with it the curvature singularity at $r = 0$—as features of the geometry rather than of the chart. *The construction is faithful as a description from its vantage; the error is ontological* [3].

2 What is derived from the object

With the substrate fixed, the construction's working object is a *cut* of it, and one instrument generates the cuts. This section states what that instrument returns, what it cannot reach, and where its boundary falls — and it separates, at each step, the theorems from the reading they ground.

2.1 Straight cuts are vacuum

In the construction's gauge the vacuum condition $T_{\mu\nu} = 0$ is not a system to be solved but a single first-order linear ordinary differential equation, $rf' + f - 1 + \Lambda r^2 = 0$, whose *entire* solution space is

$$f = 1 - \frac{2M}{r} - \frac{\Lambda r^2}{3}, \tag{1}$$

the whole Schwarzschild–de Sitter family, with M the single constant of integration [8]. *The vacuum sector is exactly the kernel of the matter functional, derived rather than matched*: straight cuts are vacuum. Taken with the bend identity below — which shows every density is realised by some cut, so the functional is onto — the two give a kernel of dimension one over a vanishing cokernel, an index of 1, generated by that constant.

Two things follow that are worth having early. There is no room in the vacuum family for a third term: the solution space is the whole of it, so a rate written on this kernel carries Λ and the cut's offset and nothing else. And the question this answers is not a new one — it is the question the programme's own 2012 dissertation left open, asking what, if not the world-matter, fixes the geometry; the kernel is the answer, and it is one the corpus states in the asking author's terms.

2.2 Matter is the bend of the cut

Writing any curve as a departure from the vacuum profile, $f = 1 - 2m(r)/r - \Lambda r^2/3$, gives $8\pi T^t{}_t = -2m'(r)/r^2$: the energy density $\rho = m'(r)/4\pi r^2$ *is the radial growth-rate of the enclosed mass*, the bend of the curve off the constant-M profile [8]. Checked on a non-vacuum member, Reissner–Nordström–de Sitter is the bend $2m(r) = 2M - q^2/r$, returning the electromagnetic stress-energy off its q^2/r^2 term.

And the reading is not an artefact of the symmetry, which is the objection it would otherwise invite. For a general leaf the density is the leaf's intrinsic-curvature departure from the substrate, $16\pi\rho = {}^3R + K^2 - K_{ij}K^{ij} - 2\Lambda$ — *which is the Hamiltonian constraint*, the spherical identity being its symmetric case [8].

What the identity states is what the bend ***is****, not why a leaf bends as it does.* That is the matter content's own generative law — a dynamics for the curve itself, as against the ordinary leaf evolution that carries it — and the construction does not supply one. *The source names it as the deepest question the construction opens onto* [8], and it is carried on the frontier below.

2.3 What the instrument reaches, and the wall

The reach is a theorem: *the range of the operator is the symmetry-reducible sector of general relativity* [6]. The size of what is reachable is set by how much symmetry a class spends — a finite parameter family where the class reduces to ordinary differential equations, a functional family where it remains a partial differential equation problem — and the reachable sector is the constrained, non-radiative skeleton of the theory.

The boundary of that reach has a positive identity, and stating it the obvious way gets it wrong. The wall is the loss of isometry, and what lies past it is *the onset of free gravitational radiation*: the graviton's two propagating polarisations, read on the matter side as dynamical inhomogeneous sources [6]. *So the wall is a seam and not a defect* — it is where generation by symmetry hands off to ordinary inhomogeneous evolution, and the handoff is a feature of the classification rather than a failure of the instrument.

One thing this section does not rest on is worth naming, because it is the part that looks like the result and is not. Within a reachable class the operator's four data — leaf, lapse, shift, vantage — supply exactly the functions the general invariant metric admits, so the cut spans the class. *P9 says plainly that this surjectivity is not the content*: once the cut carries the class's function count, the cut ansatz is the general invariant metric and spanning is near-tautological. *The content is the identification of the vacuum members as the substrate's own family — the kernel — and of matter as the bend.*

2.4 General relativity's constraint algebra is the substrate's grading

At the symmetric cut the substrate's algebra splits as $\mathfrak{so}(5,1) = \mathfrak{h} \oplus \mathfrak{m}$ with $\mathfrak{h} = \mathfrak{so}(4,1)$, and all three symmetric-space inclusions hold, $\mathfrak{m}$ not being a subalgebra. Under the correspondence between $\mathfrak{m}$ and the normal deformations and between $\mathfrak{h}$ and the tangential ones, *these are the hypersurface-deformation brackets term for term* [7].

So the algebraic shape that makes the Dirac algebra puzzling — two normal deformations bracketing into a tangential one — is exactly the shape of a symmetric-space coset: two coset directions bracketing into the isotropy. They are the same grading. *The claim is a recognition rather than an addition*: no structure is added to general relativity, and what is exhibited is that a pattern the theory already carries is the substrate's.

The coefficient that makes the algebra an algebroid rather than an algebra — the structure *function* standardly identified as the canonical root of the problem of time — is, on the symmetry-reducible reduction, the coset metric of that symmetric space [7]. *And the paper states at once that this is not a naive tensor equality*: the structure function is the Riemannian inverse spatial three-metric, the coset metric the Lorentzian five-dimensional form of signature $(1,4)$, and what is identified is the *reduced* structure function on the symmetric-cut pattern with that coset form.

2.5 What is a theorem here and what is the reading

The construction's covariance claim is that general relativity holds *a single geometry* invariant under change of chart, while this construction holds *the substrate* invariant under change of *geometry*, with the slicing curve as the gauge object: *"many line elements, one geometry" becomes "many geometries, one substrate."*

That is a reading, and the source labels it as one. The theorems — the vacuum kernel, the bend-density identity, the lapse split, the cosmological geodesic, the second-ruling embedding — are computed and verified; *the covariance reading is adopted on their strength and explicitly does not follow from them as a corollary follows from a theorem* [8]. It is stated here at that weight and at no other.

3 What is computed, and what it is confronted with

The construction meets data in three places: a rate, a set of light-element abundances, and the microwave background's angular structure. Each is stated here with the register its source gives it, and the third includes a measured disagreement for which no mechanism is in hand.

3.1 The rate, and what its two parameters are

The rate's two parameters are the substrate's curvature radius and the *offset of the cut*, and *neither is a content of the universe* [9]. The offset is measured directly and calibration-free — without a distance ladder, without the microwave background, and without any density.

The familiar cosmological form is then a coincidence of form, and the distinction matters more than it first appears. Written in the fitted pair, the same rate reads as the Friedmann one, with the density parameters following identically. *That is a translation between parameter sets and not a decomposition into components*: the two readings agree on the function and share nothing beneath it. A reader who takes the agreement as a decomposition has imported the very thing the construction declines to assume.

3.2 Three levels, and why a mis-assignment is fatal

Every rate-bearing quantity in the construction sits on one of three levels, and the assignment is not a convention [9]. *The foliation stacking rate* is read leftward — the cut is primary and a density is the name of its bend rather than its cause — and it is empirically forced rather than modelled. *The leaf-level local dynamics* is the ordinary Friedmann readout with radiation gravitating normally: the same single geodesic, read *inward* as dust collapse where the first reads it *outward* as cosmology. *The projection* is what an observer reads.

There is no boundary in time between the first two levels. The distinction is kinematic and holds at every epoch, the decomposition exact, the two rates differing by the radiation term alone. *A quantity assigned to the wrong level is not approximately right*; it is a different physical claim, and §3.5 records a case where the corpus itself made that error and what it cost.

3.3 The abundances, and the miss that discriminates

The cooling leg *is* a standard big-bang nucleosynthesis — cooling through the window at the standard rate from a fully dissociated start — and a genuine multi-nuclide network integrated explicitly on that history returns $Y_p = 0.247$ against an observed 0.245, $D/H = 2.51 \times 10^{-5}$ against 2.53×10^{-5}, ^{3}He at 1.05×10^{-5}, and ^{7}Li at the standard several-fold over-prediction — *all jointly from the single inherited η*, at the baryon density the microwave background reads from the peak heights, with deuterium at -0.5σ and helium-4 at $+0.5\sigma$ [13].

And the lithium miss is the discriminating result rather than an embarrassment to be noted and passed over. It is the standard over-prediction, reproduced — which is to say the construction inherits the standard problem exactly, neither curing it nor worsening it. *A mechanism that changed the early history would have moved lithium*; this one does not, because the window and the rate through it are the standard ones.

3.4 The low-multipole floor, and why it does not discriminate

Because the distance slicing is exactly flat while the cosmological layers are a closed three-sphere, the photons are projected through the *flat* geometry while only the *source* modes carry the closed quantisation — *not the hyperspherical transfer of a literal closed universe*, which would deliver the lowest mode to the quadrupole and no deficit at all. The result is a *parameter-free* deficit below $\ell \approx 8$, bottoming at $\ell = 4$ and *not* at the quadrupole [9].

The register here is split and the split is the honest part: the existence, location and minimum are established; *the depth is open*, two transfers differing by up to a factor of two. And the feature does not discriminate between frameworks — it is a prediction the construction makes for free, not a test it passes and a rival fails.

3.5 The acoustic comparison, and a deficit with no mechanism

The acoustic *scale* is met, and the corpus records what meeting it cost: it is an accommodation, and it is spent. *The peak structure is a separate confrontation and it does not come out even.*

Run at converged wavenumber on the leaf congruence — the rate the level rule assigns the perturbations to — with the instrument calibrated in the same configuration against a standard code, the construction returns a first peak at $\ell_1 = 204$ and $\ell_1/\ell_A = 0.6764$ against the sky's 0.7312: *a 7.5% deficit in position*, with height ratios $P_1/P_2 = 1.759$ and $P_1/P_3 = 1.612$ against 2.217 and 2.277. *Two wavenumber cutoffs agree to every digit, and the deficit is neither of the two instrument faults* that a reader would reasonably suspect — the truncation and the projection path are both removed in that configuration.

The seam datum's own freedoms are measured rather than left as an escape. Across the seventeen readings admitted by a four-peak criterion fixed before the numbers, the first peak spans 148 to 228 — a factor of 1.541, and 1.118 across the numeric-phase readings alone —

so *the position is a statement of the construction and not a free choice*, and the sky's value lies inside that span. The same holds separately for the spacing, the phase intercept and the alternation ratio. *But a span containing the sky on each statistic one at a time is much weaker than a reading that reproduces the sky, and there is no such reading.*

What the readings do carry is a correlation, and it is the sharpest thing the scan returns. Position and alternation move against each other across the datum, at Spearman $\rho = +0.782$ and $p = 2.1 \times 10^{-4}$: as the datum carries the peak up toward the observed value, the second gap turns from contracting to expanding. Of the readings landing the first peak within one grid step of the sky's, none contracts; of those that contract, every one sits at $\ell_1 \leq 212$. *So the seam datum can buy the position or the alternation and not both, and no mechanism for that trade is in hand.*

This is where the level rule earns its statement, because the deficit's most natural explanation turns on it. It is tempting to say that the acoustic modes re-enter above the plasma's onset, so that none of them is driven and the comb should be uniform for that reason. *That census is taken on the stacking rate, and the rule assigns the perturbations to the leaf*, which carries a radiation term: on the leaf the band containing the first peak enters the horizon *while radiation dominates*. Those modes are driven, and the driving's size is measured — removing it moves the first peak by 136 multipoles.

So the deficit stands as a measurement without a mechanism. The construction's own level assignment rules out the explanation that would have covered it, and what remains is a number, its size, and the two places it is not: not the wavenumber truncation, and not the projection path.

References

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[12] D. Janzen, *The matter boundary: what the substrate's symmetry can and cannot carry.*

[13] D. Janzen, *Cosmogenesis: the branch point, the abundances, and what the crossing carries.*

[14] D. Janzen, *The geometric core: the substrate as a real object, and maximal symmetry worn seven ways.*

Appendix L The Knowledge Ledgers

Each claim marked $^{\mathsf{L}}$ in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

`QUADRIC_GEOMETRY_LEDGER.md` [field bake]

projective geometry of quadrics against the CR substrate — the CK metric identification, and the ladder gap it exposes